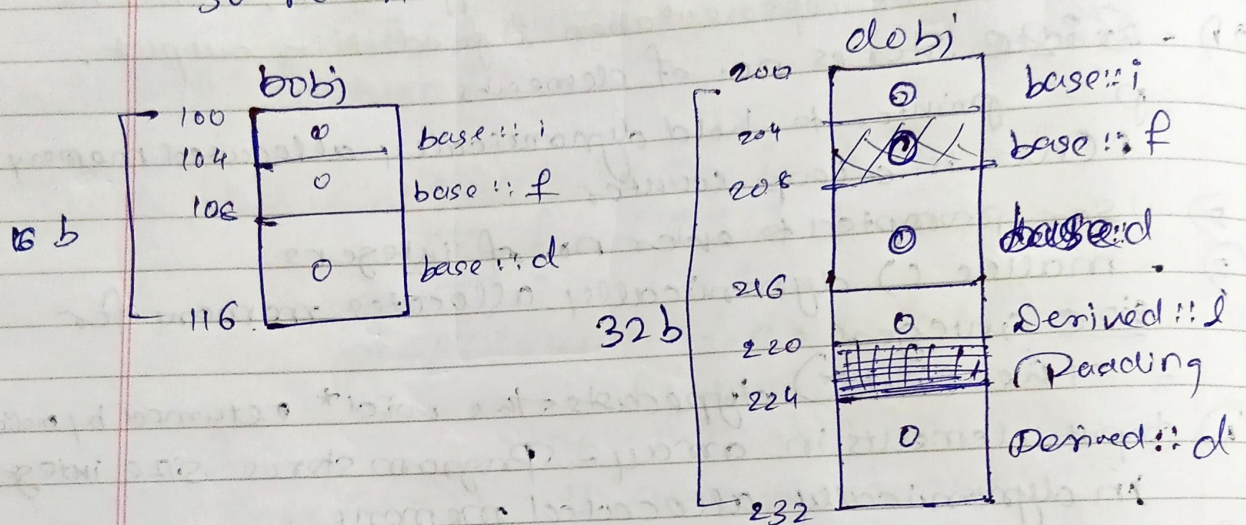


Assignment - 20

1. Draw object layout of below code & explain the internal working in detail.

- 7
- Derived class inherits from base publicly
 - Derived defines its own `i` & `d`, which shadow base's `i` & `d`.
 - `fun()` is overridden in Derived, but not virtual so no vtable is created.



2. Memory Arrangement:

- Derived inherits base so its objects (`dobj`) contain base's members first.
- Derived's members are stored after base's.

3. Member Shadowing:

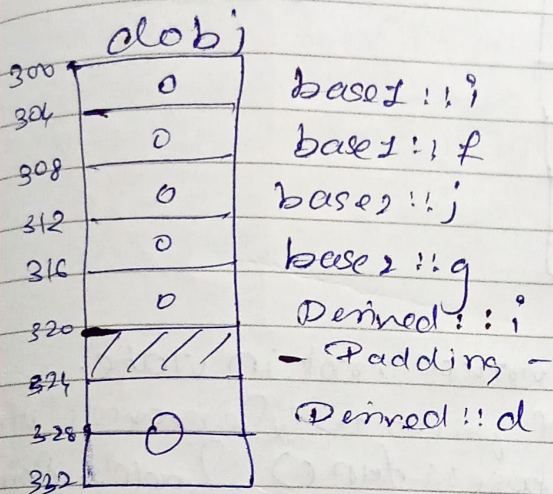
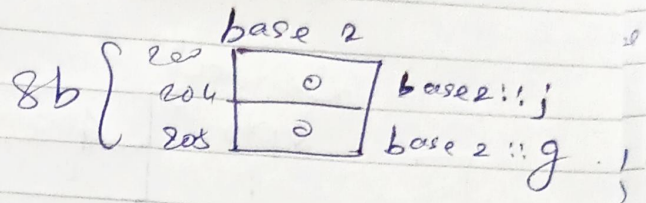
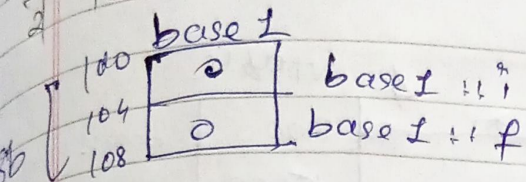
- Derived redefines `int i` & `double d`, hiding base's `i` & `d`.

4. Function overriding:

- `fun()` is redefined in Derived, so:
 - `bobj.fun()` calls `base::fun()`
 - `dobj.fun()` calls `Derived::fun()`

4. Padding & Alignment
- `int d (derived)` (4 b) is followed by 4 bytes padding
 - `double d (derived)` (8 b) is aligned properly.

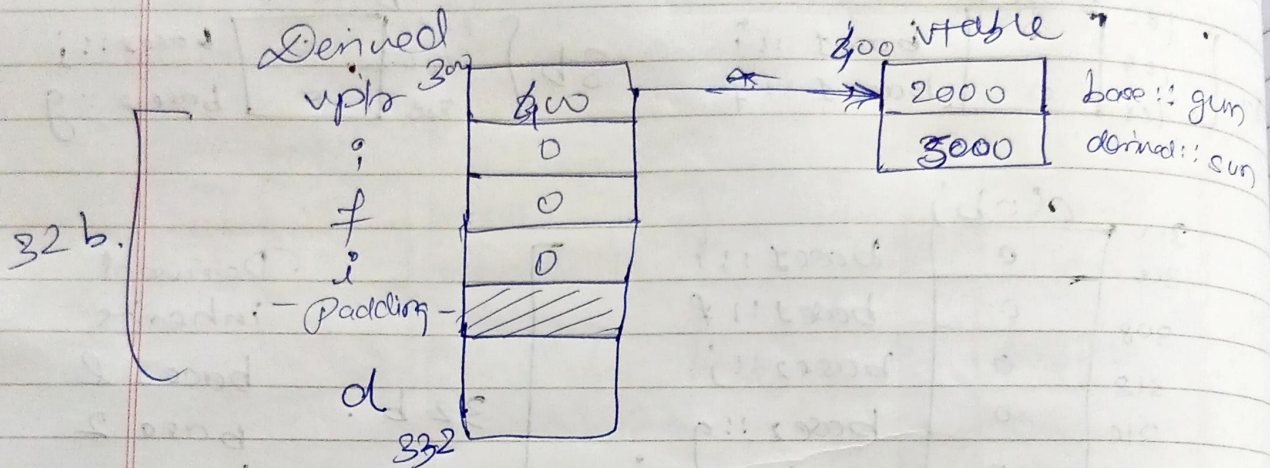
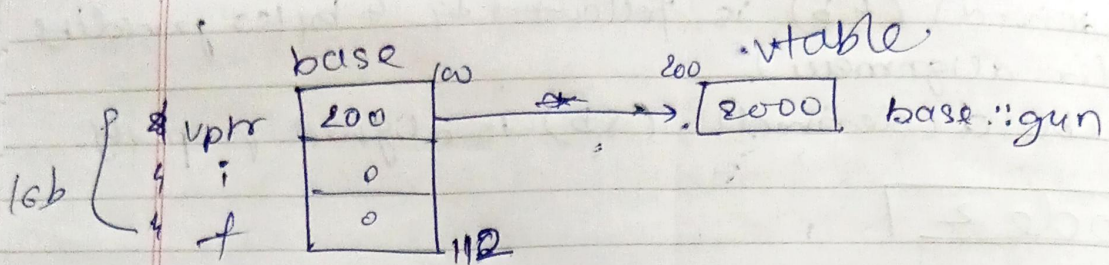
Code 2



Derived inherits
base 1 &
base 2
i.e.
'multiple Inheritance'

- 1) Derived contains both base 1 and base 2 subobjects.
- 2) Derived defines `i`, which hides base 1's `i`.
- 3) `func()` in derived hides `func()` in base 2, but it's not overridden since `func()` in base 2 is not virtual.
- 4) `Derived :: func()` hides `base :: func()` but does not override since `base 2 :: func()` is not virtual.
- 5) `Derived :: func()` is a new function in Derived (not affecting base classes).
- 6) To align `double d`, 4 bytes of padding is added after `int i` (Derived).
- 7) `double d` is stored at an 8-byte aligned location for efficiency.

3] Code 3



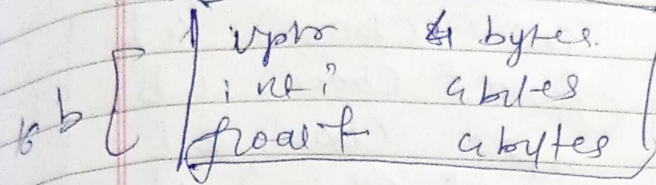
fun(c) - Non-virtual, not in vtable.
vptr in Derived points to Derived's vtable, which overrides base::fun(c) & adds Derived::sun.
Derived::gun(c) hides base::gun(c), but since base::gun(c) was virtual, Derived does not override it in a value.

4] Code 4

- Derived publicly inherits from base; where base has 2 virtual functions (gun(c), sun(c)) & 2 non-virtual functions (fun(c), run(c)).

Derived:
overrides fun(c), gun(c) & run(c) (making run(c) virtual, hides sun(c) by redefining it (but does not override) virtual function generate vtable & vptr.

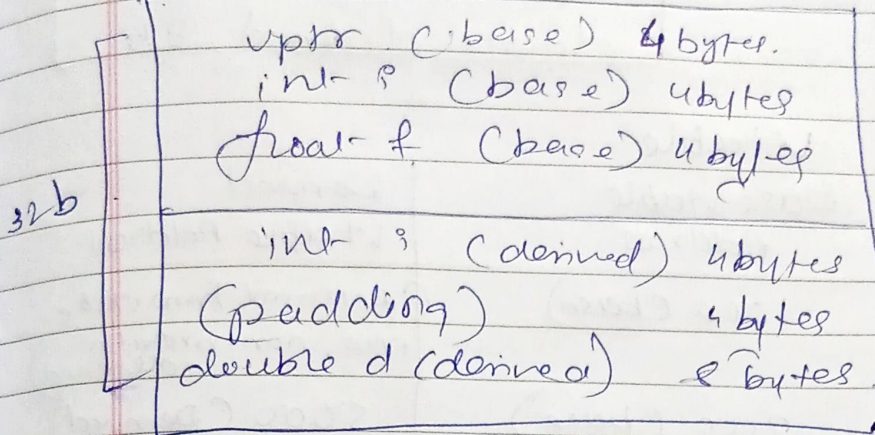
base object



vtable

2000	base: gun
3000	base: sun

Derived object



vtable

6000	derived: gun
3000	base: sun
5000	derived: func
8000	derived: runc

vtable

Function	Base vtable Address	Derived vtable Address
gun()	2000 (base)	6000 (derived)
sun()	3000 (base)	3000 (base)
fun()	(Non-virtual)	5000 (derived)
runc()	(Non-virtual)	8000 (derived)

5] Code 5

- Derived publicly inherits from base, where:
 - base has 2 virtual functions (gun(), sun())
- Derived introduces:
 - A new virtual function - fun()
 - A non-virtual redefⁿ of - gun()
 - An override of sun()

base object		
int i	4 b	
float f	4 b	
vptr	4 b	

Derived		
int i (base)	4 b	
float f (base)	4 b	
vptr (base)	4 b	
int i (derived)	4 b	
padding	4 b	
double d (derived)	8 b	

Function	Vtable Base vtable Address	Derived Vtable Address
gun()	1000 (base)	(removed from vtable, now, non-virtual in derived)
sun()	2000 (base)	5000 (Derived)
fun()	(Not present in base)	3000 (Derived)

- fun() is new in Derived, so it appears only in Derived's vtable.
- gun() in Derived is non-virtual, so it does not override base::gun(), meaning Derived's vtable does not contain gun(). Calls to gun() in Derived are statically bound.
- sun() in Derived overrides base::sun(), so Derived's vtable replaces base::sun() with its own version. vtable in Derived includes overridden sun() and new fun() but removes gun().